

Analysis

$$\begin{cases} dX_t^i &= P v(\theta_t^i) dt + \sqrt{2\sigma_x} dW_t^i \\ d\theta_t^i &= \chi n(\theta_t^i) \cdot \frac{1}{N} \sum_{j \neq i} \nabla K(X_t^i + \tau v(\theta_t^i) - X_t^j) + \sqrt{2} dB_t^i \end{cases}$$

- Well-posedness
 - Singular (Newtonian) ∇K : $|\nabla K(x)| \sim |x|^{-1}$
 - For $\sigma_x > 0$, **if τ small enough**, solutions exist for all N and any time $t \geq 0$
 - And: convergence in law to mean-field limit as $N \rightarrow \infty$ ($L^1 \cap L^\infty$ chaotic initial law)
 - Via recent compactness method for Keller-Segel (Jourdain & Fournier, Ann. Appl. Prob., 2017) (see dW, thesis, 2025)

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